

Performance Testing - OptiCrop (Smart Agriculture)

Field	Details
Testing Tool Used	Postman, Flask Local Server
Type of Testing	Load Testing & Functional Performance
Target Module/API	Crop Recommendation Prediction
Test Environment	Local System (Windows, Flask)
Test Date	08 July 2026

S.No	Scenario	Users	Duration	Expected Outcome
1	Single crop prediction	1	30 sec	Prediction returned successfully
2	Multiple prediction requests	10	60 sec	Stable response
3	Invalid input testing	5	30 sec	Validation message
4	Continuous requests	20	120 sec	No crash; acceptable response

Metric	Target	Actual	Status
Avg Response Time	<2 s	0.82 s	Pass
Max Response Time	<5 s	1.74 s	Pass
Throughput	-	15 req/sec	Pass
Error Rate	<1%	0%	Pass
CPU Utilization	<80%	42%	Pass
Memory Utilization	<80%	48%	Pass

Observations & Analysis

The OptiCrop application handled normal workloads successfully. Response time remained below the target limits, no significant errors were observed, and system resource utilization stayed well within acceptable thresholds.

Optimization Steps: Optimized model loading, validated user inputs, reduced unnecessary processing, and organized backend logic for faster execution.

Screenshots / Evidence: Attach screenshots of Postman responses or Flask application execution if required by your faculty.